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Topic Modeling

Natural Language Processing

Prodi Rekayasa Kecerdasan Artifisial

Departemen Teknik Informatika

Motivation

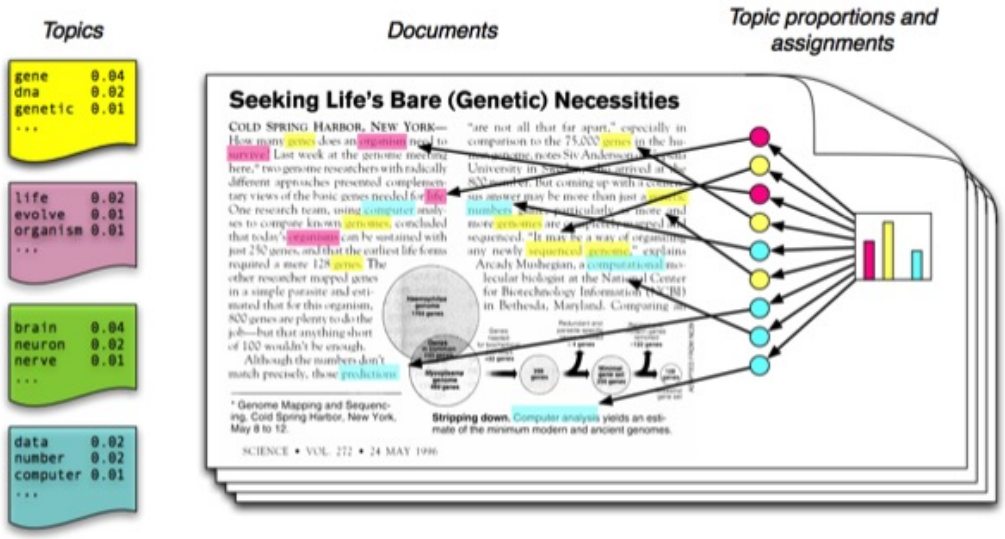


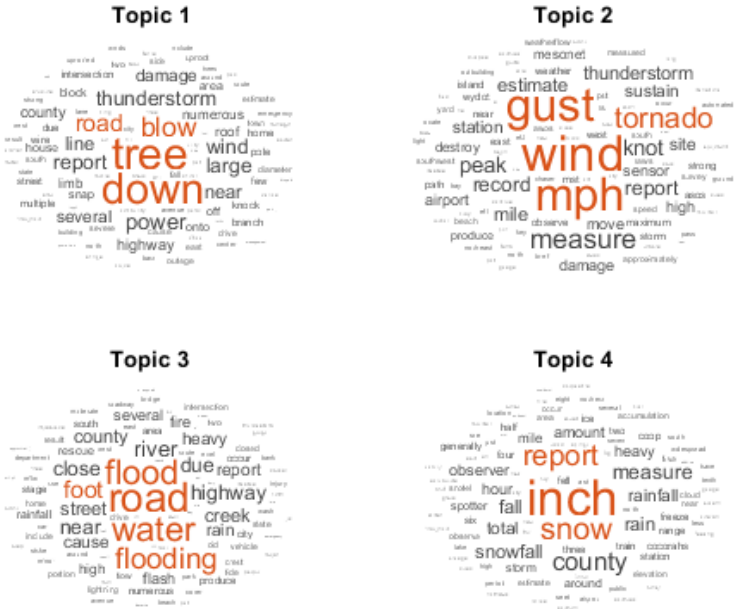
Figure source: Blei, D. M. (2012). Probabilistic topic models. *Communications of the ACM*, *55*(4), 77-84.

- Massive documents => difficult to organize, summarize and understand
- Each topic is a distribution of words; each document is a mixture of corpus-wide topics; and each word is drawn from one of those topics
- The goal is to infer the hidden variables from the documents

What is Topic Modeling

Unsupervised learning technique for identifying hidden topics from a collection of text data

It was done by analyzing word patterns and grouping similar documents together



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Topic Modeling vs Clustering

Topic Modeling

- Focuses on discovering hidden themes in text data.
- A document can belong to **multiple topics** with varying probabilities
- Output: Topics represented by word distributions.
- Algorithms: LDA, NMF, BERTopic.

Clustering

- Groups similar documents into distinct clusters.
- A document belongs to only **one** distinct cluster.
- Output: Groups (clusters) of similar data points.
- Algorithms: K-Means, Hierarchical Clustering, DBSCAN.

Basically, topic modeling captures **semantic themes**, clustering captures **document similarity**

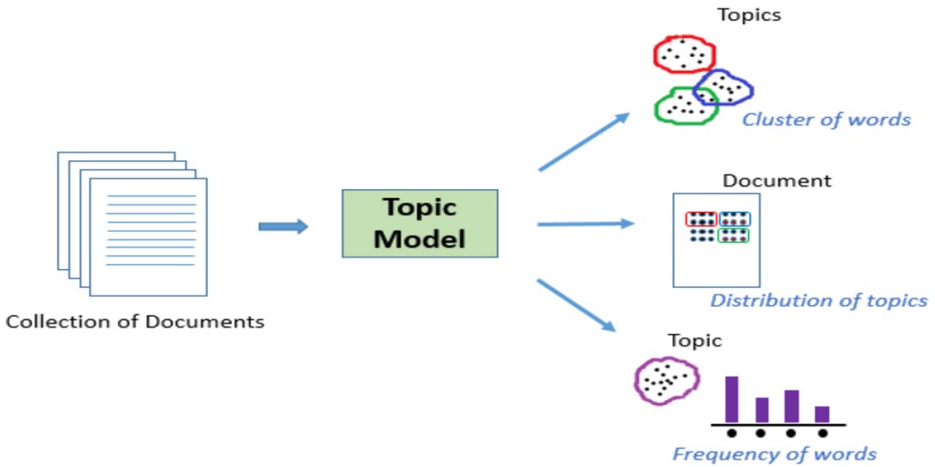
How topic modeling works?

Input Data: Topic models take a collection of documents (like articles, reviews, or tweets) as input.

Word Frequency: The model analyzes the frequency of words in each document and across the entire dataset.

Topic Discovery: It identifies groups of words that often appear together, forming "topics." For example, words like "bat," "ball," and "score" might form a topic related to sports.

Probabilistic Approach: Popular algorithms like Latent Dirichlet Allocation (LDA) use probability to assign each document a mix of topics and each topic a mix of words.

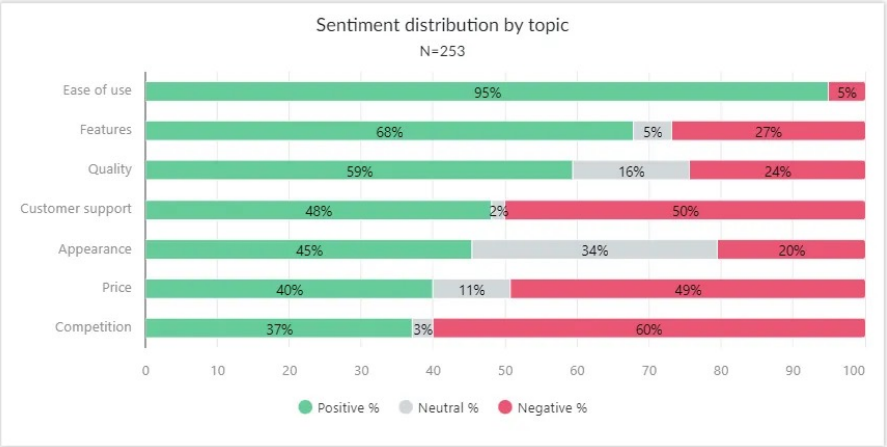


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Use case of Topic Modeling

just now	1 hour ago	1 hour ago
1 #MARK_1999 488K	1 #MARK_1999 483K	1 #MARK_1999 477K
2 #마크의_1999_솔로데뷔 253K	2 #마크의_1999_솔로데뷔 249K	2 #마크의_1999_솔로데뷔 246K
3 #13YearsWithEXO 183K	3 #13YearsWithEXO 175K	3 #13YearsWithEXO 165K
4 #EXO13thAnniversary 159K	4 #EXO13thAnniversary 151K	4 #EXO13thAnniversary 142K
5 #WeAreOneandThirteen 132K	5 #WeAreOneandThirteen 127K	5 #WeAreOneandThirteen 120K
6 IHSG 22K	6 IHSG 17K	6 serangcerdas tolaknepotisme
7 13 YEARS ON EXO PLANET 63K	7 serangcerdas tolaknepotisme	7 13 YEARS ON EXO PLANET 59K
8 ratunepotisme bukanpilihan	8 Trading Halt	8 ratunepotisme bukanpilihan
9 sehun 21K	9 13 YEARS ON EXO PLANET 61K	9 YG'S SILENCE 41K
10 Trading Halt	10 sehun	10 JEONGW

View all 50 trends



- Trending topic
- Topic modeling for sentiment analysis
- Document annotation

Topic Modeling Algorithms

Latent Dirichlet Allocation (LDA)

- A probabilistic model that represents documents as mixtures of topics.

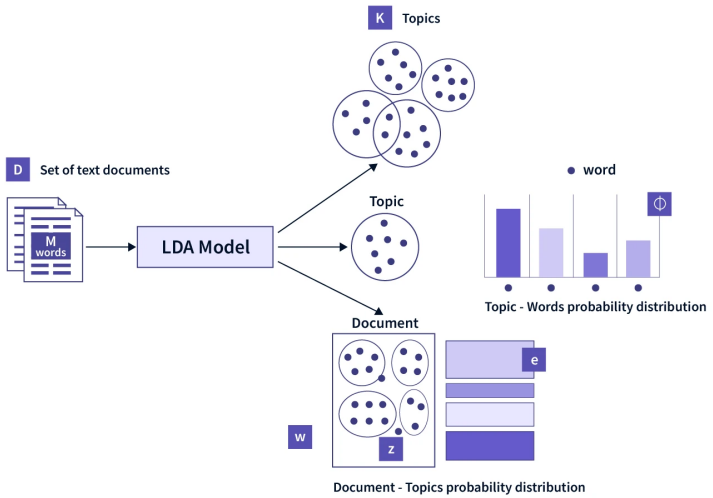
Non-Negative Matrix Factorization (NMF)

- A linear algebra-based approach that decomposes a document-term matrix into two interpretable matrices.

BERTopic (BERT-based Topic Modeling)

- A deep learning-based approach leveraging transformer embeddings and clustering.

Latent Dirichlet Allocation (LDA)



SCALER Topics

LDA is a generative Bayesian model for topic modeling

Assumptions on all LDA variables:

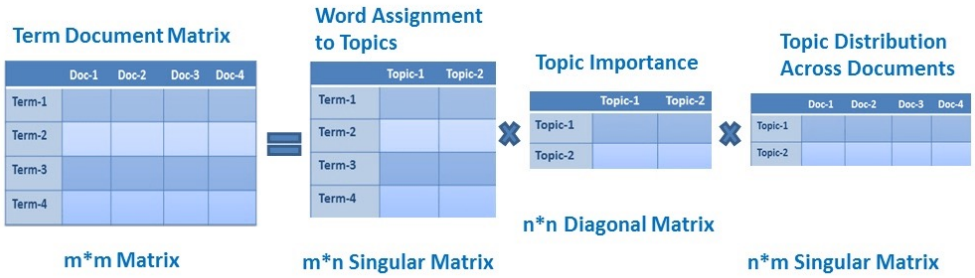
Word: the basic unit of discrete data

Document: a collection of words (exchangeability assumption)

Corpus: a collection of documents

Topic (hidden): a distribution over words & the number of topics K is known.

Latent Semantic Analysis (LSA)



The core idea is to take a matrix of documents and terms and try to decompose it into separate two matrices – a document-topic matrix and a topic-term matrix.

Steps:

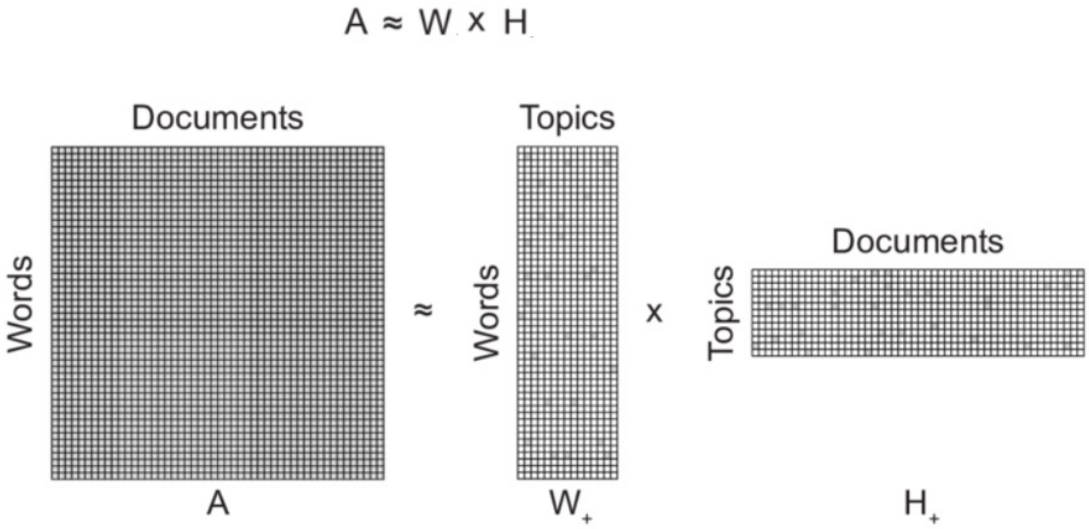
1. Generate a document-term matrix of shape m x n
2. Perform dimensionality reduction using Singular Value Decomposition (SVD)

U matrix: In this matrix, rows represent document vectors expressed in terms of topics.

V matrix: In this matrix, rows represent term vectors expressed in terms of topics.

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Non-negative Matrix Factorization (NMF)



NMF is a topic modeling technique that decomposes a document-term matrix into two non-negative matrices: a document-topic matrix and a topic-term matrix, revealing latent topics within a collection of documents

How it works:

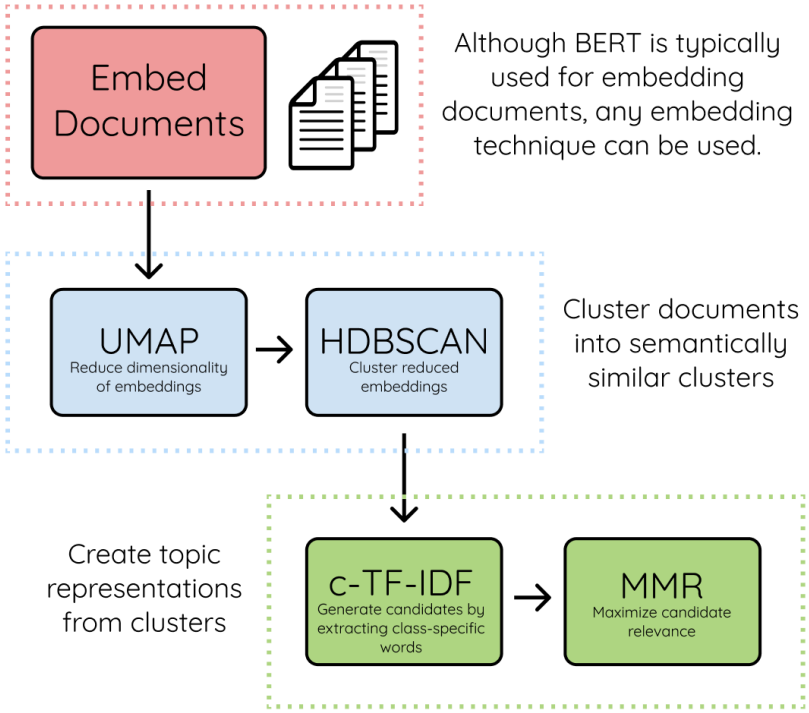
Input: NMF takes a document-term matrix (where rows represent documents and columns represent terms) as input.

It decomposes this matrix into two lower-dimensional matrices:

- Document-Topic Matrix (W): Rows represent documents, and columns represent topics. The entries in this matrix indicate the strength of each topic in each document.
- Topic-Term Matrix (H): Rows represent topics, and columns represent terms. The entries in this matrix indicate the terms that characterize each topic.

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BERTopic



BERTopic leverages **BERT embeddings** and the concept of **c-TF-IDF** (class-based TF-IDF) to create coherent and easily interpretable topics, described by automatically generated labels

Steps:

1. **Document Embedding:** first we need to create document embeddings
2. **Document Clustering:** dimensionality reduction through UMAP, then apply a density-based clustering technique, such as HDBSCAN
3. **Topic Representation:** to represent each topic (i.e. cluster of embeddings), we can modify the well-known TF-IDF score. Then extract the most important words per cluster to get descriptions of topics (**class-based TF-IDF**)

Algorithms Comparison

Feature	LDA	LSA	NMF	BERTopic
Type	Probabilistic	Linear algebra	Matrix factorization	Embedding-based
Output	Topic probabilities	Latent vectors	Topic weights	Clusters + keywords
Requires TF-IDF?	No	Yes	Yes	No (uses BERT embeddings)
Interpretability	High	Medium	High	Very high
Speed	Moderate	Fast	Moderate	Slower (deep models)
Best for	Long documents	Semantic structure	Interpretable topics	Context-rich text

Model Evaluation

Perplexity

Perplexity measures the modeling power by calculating the inverse log- likelihood of unobserved documents

Better models have lower perplexity, suggesting less uncertainties about the unobserved document

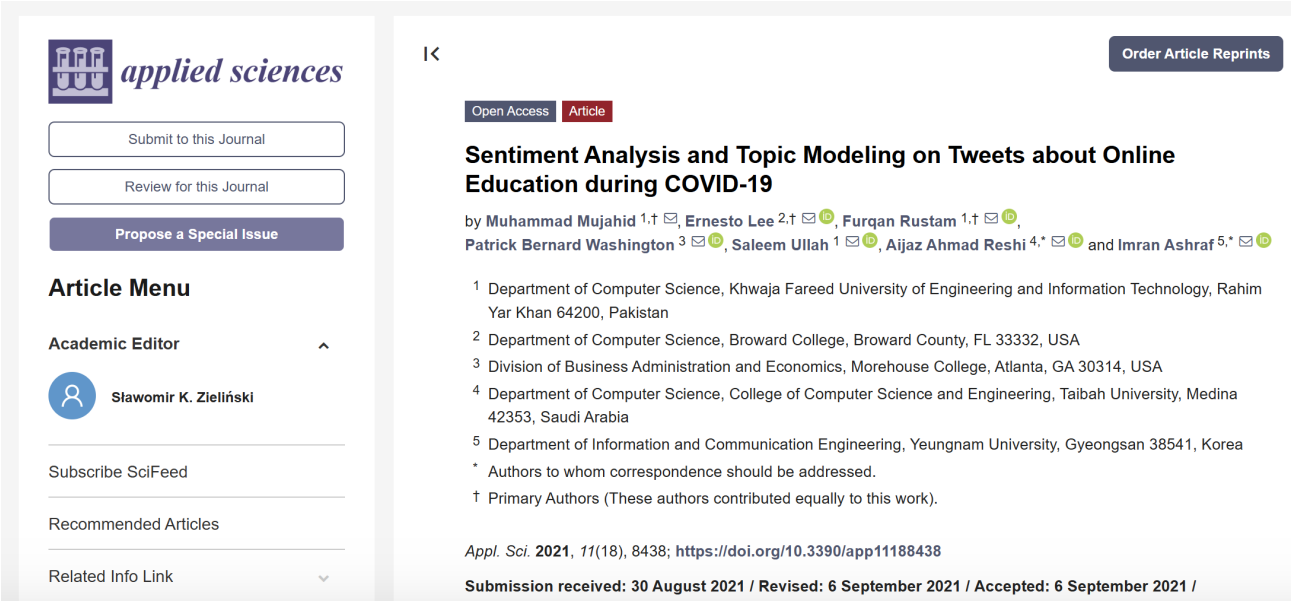
Topic coherence

It measures the semantic similarity among the top keywords of a topic

A high coherence score (closer to 1) suggests that the words in the topic are highly related and form a coherent unit

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Sentiment Analysis and Topic Modeling on Tweets about Online Education during COVID-19

by Muhammad Mujahid [1,†], Ernesto Lee [2,†], Furqan Rustam [1,†], Patrick Bernard Washington [3], Saleem Ullah [1], Aijaz Ahmad Reshi [4,*] and Imran Ashraf [5,*]

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Appl. Sci. **2021**, *11*(18), 8438; **https://doi.org/10.3390/app11188438**

Submission received: 30 August 2021 / Revised: 6 September 2021 / Accepted: 6 September 2021 /

It investigates public opinions on the shift to online education during the COVID-19 pandemic by analyzing 17,155 tweets. The researchers employed both machine learning and deep learning techniques to assess sentiment and identify prevalent topics

Through **Latent Semantic Analysis (LSA)**, the study uncovered primary concerns related to online education

https://www.mdpi.com/2076-3417/11/18/8438

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Recent Researches



frontiers | Frontiers in Sociology

METHODS
published: 06 May 2022
doi: 10.3389/fsoc.2022.886498

Check for updates

A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts

Roman Egger[1] *and Joanne Yu*[2*]

[1] *Innovation and Management in Tourism, Salzburg University of Applied Sciences, Salzburg, Austria,* [2] *Department of Tourism and Service Management, Modul University Vienna, Vienna, Austria*

https://www.frontiersin.org/journals/sociology/articles/10.3389/fsoc.2022.886498/full

It evaluates the effectiveness of four topic modeling techniques—Latent Dirichlet Allocation (LDA), Non-negative Matrix Factorization (NMF), Top2Vec, and BERTopic—in analyzing Twitter data related to COVID-19 travel discussions

The research suggests that while traditional models like LDA and NMF are useful, embedding-based models such as BERTopic offer enhanced performance in uncovering detailed and contextually rich topics from short, unstructured text data like tweets

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Recent Researches

Information Systems 112 (2023) 102131

Contents lists available at ScienceDirect

Information Systems

journal homepage: www.elsevier.com/locate/is



Topic modeling algorithms and applications: A survey

Aly Abdelrazek [a,*], Yomna Eid [a], Eman Gawish [a], Walaa Medhat [a,b], Ahmed Hassan [a]

[a] Information Technology and Computer Science, CIS, Nile University, Giza, Egypt
[b] Faculty of computers and artificial intelligence, Benha University, Egypt

The paper organizes topic modeling methods into four primary categories: Algebraic, Probabilistic, Neural Network-based, and Hybrid method

This also discuss several evaluation metric to assess topic modeling algorithms

https://www.sciencedirect.com/science/article/pii/S0306437922001090#sec5

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Thank you